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Beauty Creations

Scientific Brochure

Hyalurosmooth™

The mindful way to cope with everyday stress

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We create chemistry

Hyalurosmooth™

The mindful way to cope with everyday stress

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SUMMARY FILE

Hyalurosmooth™*	PW LS 8997 (POWDER FORM) - LS 8998 (LIQUID FORM)
Origin - Description	Natural galactomannans from the seed of the Ayurveda plant *Cassia angustifolia*
Regulatory data	
INCI	PW LS 8997: Cassia Angustifolia Seed Polysaccharide LS 8998: Water (and) Cassia Angustifolia Seed Polysaccharide (and) Hydroxyethylcellulose
China	Each component is listed in “Inventory of Existing Cosmetic Ingredient in China” (IECIC 2015)
CAS#	PW LS 8997: 85187-05-9 LS 8998: 7732-18-5; 85187-05-9; 9004-62-4
EINECS#	PW LS 8997: 286-092-5 LS 8998: 231-791-2; 286-092-5
Appearance	PW LS 8997: White to beige powder LS 8998: Light beige fluid gel
Preservative	PW LS 8997: None LS 8998: Phenoxyethanol
Natural labels	None
Cosmetic uses	
Properties	• Mindfulness • Hydration on stressed skin • Radiance/Gloss on stressed skin • Appearance of skin plumping, moisturizing and long lasting • Moisturizing body care to reduce ashiness • Botanical alternative to hyaluronic acid or sodium hyaluronate
Applications	Increase hydration and Radiance (Gloss) on stressed skin Day and night rejuvenating serum
Formulation data	
Concentration of use	PW LS 8997: 0.05%-0.1% LS 8998: 2.5-5%
Solubility	Soluble in water
Incorporation method	PW LS8997: can be incorporated into the hot water phase of emulsions up to 80°C or dissolved into water at room temperature for cold processing LS 8998 : can be incorporated into the finished process below 50°C, or at room temperature for cold processing
Optimal pH	PW LS 8997 and LS 8998: between 4 and 9
Manufacturer	BASF Beauty Care Solutions France SAS 3, Rue de Seichamps 54425 Pulnoy

* This product does not contain hyaluronic acid (or derivatives)



OUR SKIN UNCONSCIOUSLY EXPRESSES THE EMOTIONS OF THE RIGHT SIDE OF OUR BRAIN

Mindfulness

Finding its roots in Eastern contemplative traditions, Mindfulness is called the "heart" of Buddhist meditation and is mostly associated with the formal practice of mindfulness meditation[1]. **Mindfulness** is a process of consciously attending and awareness to the expression of the present moment. It helps us avoid running on autopilot, to maintain control and awareness over our lives[2,3]. The practice of **Mindfulness** is employed in psychology to reduce stress and anxiety. By better managing our emotions, we feel better, we increase our well-being and we feel healthier[4].

Ayurveda is the oldest form of health care in the world and originated in India over 5000 years ago. Ayurveda is a science of life (Ayur = life, Veda = science or knowledge) which offers a body of wisdom designed to help people stay vibrant and healthy while realizing their full human potential. Ayurveda is defined by two main guiding principles: 1) the inextricable connection between the mind and the body, 2) the effect on the mind to heal and transform the body.

How are the skin and mind connected?

Skin is the primary sensing organ subjected to external stressors such as heat and cold. When we feel stressed, we may sometimes undergo physical tension, indicating that our mind and skin are closely linked. Moreover, our skin is also connected to our emotions: blushing with embarrassment or having goose pimples are examples of visible signs that our skin reflects what we feel inside.

It has been demonstrated that stress on students during final exams or stress due to personal problems may cause a significant decline in skin's permeability barrier and recovery kinetics[5,6]. Psychological stress during an interview increases the cortisol level which may cause a delay in the skin's barrier function recovery, and may activate several inflammatory and immune factors[7]. The body's cortisol level is regulated daily by the internal circadian clock system[8].

What is cortisol?

Cortisol, often referred to the “stress mediator”, is produced by the adrenal glands. Under normal circumstances, the body maintains a basal level of cortisol that is necessary for different body functions (Anti-inflammatory actions, heart and blood vessel tone and contraction, central nervous system activation, fat, protein and carbohydrate metabolism etc...). Cortisol reaches a peak early in the morning and falls around midnight[9,10].

In response to a stress, cortisol is secreted in higher levels and then returns to a normal level.

Unfortunately, in a stressful environment, the stress response is activated continually, without the body having the opportunity to return to a normal cortisol level.

In this case, an elevated cortisol level has negative impacts on the body in general, and on the skin in particular. Moreover, human skin exposed to a dry environment can increase the production and release of cortisol by as much as three-fold in comparison to the skin in a controlled environment[11].

How does cortisol affect the skin?

Stress can significantly up-regulate the cortisol level resulting in major impacts on skin homeostasis, hydration, inflammation and maintenance of the skin’s appearance[11,12]. Excess of cortisol induces changes in the epidermis by reducing its thickness and decreasing the size of its keratinocytes and consequently flattens the dermal epidermal junction. The decreased size and suppressed proliferation of keratinocytes may result in structural defects in the epidermis. The consequences are an increased permeability and an increased trans epidermal water loss (TEWL) which indicates a disrupted barrier function of the skin[13].
Cortisol downregulates the expression of epidermal antimicrobial peptides and increases the severity of cutaneous infections. Cortisol also inhibits wound healing[14,15,16].
Cortisol decreases the formation of connective tissue by inhibiting fibroblast proliferation and synthesis of both collagen and glycosaminoglycans, particularly hyaluronic acid[17,18,19].

Conclusion

The skin is the primary sensing organ exposed to external stressors and can show physical signs of psychological stress and emotions (blushing, paleness…). Skin reflects what we are feeling inside; and its aspect may be negatively or positively affected by emotions, stress or a mindful life style.
As the number of people suffering from psychological stress increases, the worldwide demand for mindful products to fight against these changes also grows.
At a molecular level, the accumulation of stressful situations leads to a high level of cortisol, the main stress mediator, and induces adverse effects on the skin such as a decrease of collagen or hyaluronic acid synthesis.

However, the stress related-effects on our skin’s appearance are open to question. Moreover, we pondered how to help the skin to be in a mindful state?
A mindful solution targeting these alterations could be a solution to put the skin into a mindful state.
To answer to this question, the Beauty Creations team has conducted a consumer survey, and created a solution focused on controlling both cortisol level in the skin as well as its negative effects at the molecular level to improve the skin’s appearance.

The Beauty Creations team has developed Hyalurosmooth™ to help the skin to return to and stay in a mindful state.

GALACTOMANNANS FROM CASSIA ANGUSTIFOLIA SEED



Description of the plant

Family: *Fabaceae*

Species: *Cassia angustifolia Vahl (accepted names Cassia senna L., Senna alexandrina Mill.*

Common name: Indian Senna, Alexandrian Senna, Séné (French)

Description: *Cassia angustifolia* is a fast-growing shrub. Its branches are erected and its leaves are opposite, oval, lanceolate and glabrous. The flowers are yellow and produce a fruit in the form of a greenish to dark brown pod comprising 5 to 7 oval seeds. The fruit ripen in September.

Distribution: *C. angustifolia* is native to India. It grows in semi-desert areas, low meadows, along rivers, embankments and valleys. This species does not tolerate excessive irrigation. *C. angustifolia* grows well in dry soil.

Origin: India

Plant part used: seed

Cassia angustifolia: an Ayurvedic plant

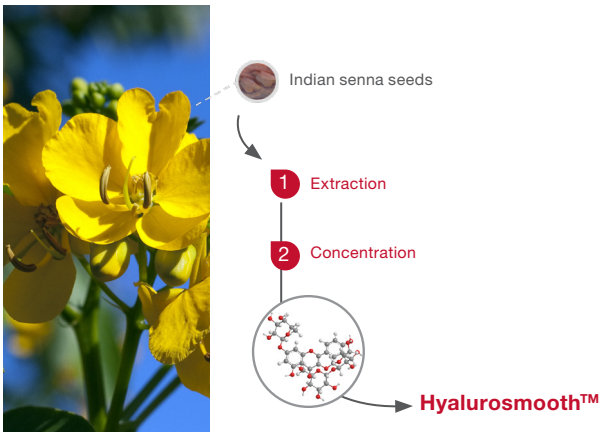
For more than 3000 years, the official ayurvedic pharmacopeia has been enriched with 1200 to 1500 plants. All plants have been studied thoroughly before becoming part of the ayurvedic pharmacopeia[20]. Cassia angustifolia is a natural Ayurvedic herb more popularly known as Senna. It is mainly used as a blood purifier, for relieving constipation and to help overcome skin diseases.

Processing and composition

Naturally rich in polysaccharides, the shrub Cassia angustifolia is an ideal source of galactomannans. Following their isolation and separation from the seeds of the Senna shrub, galactomannans are purified by appropriate enzymatic treatment.

Hyalurosmooth™ galactomannans are native botanical hetero-polysaccharides, with a high molecular weight (~ 170,000 Daltons) composed mainly of mannose (≥ 50%) and galactose (≥ 15%) monomers. Hyalurosmooth™ powder is characterized by a total carbohydrate content over 85% (Figure 1).

These polysaccharides strongly bind to water, as does hyaluronic acid, and may form due to their high molecular weight, a protective film on surfaces.



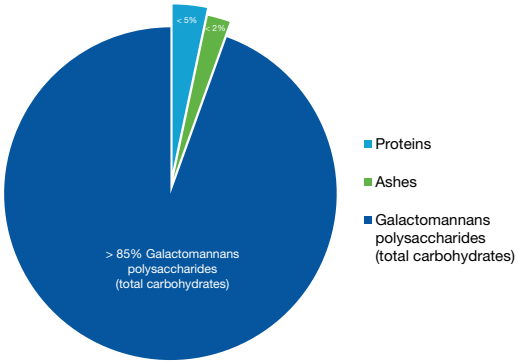


Figure 1 - Typical phytochemical composition of Hyalurosmooth™ powder.

Cassia angustifolia polysaccharides have also been described to soothe the skin by exhibiting an anti-inflammatory function and antibacterial effect[21,22].

Safety and tolerance data of the product

Hyalurosmooth™, was tested to ensure its safety in the recommended conditions of use. Hyalurosmooth™ does not irritate the eyes or skin under normal conditions of use. No indication of skin sensitization was observed.

Description

Hyalurosmooth™ is a natural, botanically derived high molecular weight polysaccharides (galactomannans) from the seeds of the Indian plant *Cassia angustifolia*.

Hyalurosmooth™ is available in **two forms**:

As a preservative free, concentrated powder (PW LS 8997).
INCI name: Cassia Angustifolia Seed Polysaccharide.

As a colloidal solute in water (with Phenoxyethanol as preservative) (LS 8998).
INCI name: Water (and) Cassia Angustifolia Seed Polysaccharide (and) Hydroxyethylcellulose

Dose of use: 0.05 - 0.1% (PW LS 8997) / 2.5 - 5% (LS 8998)



China IECIC listed

Example: Illustration of a commercial sample of Hyalurosmooth™ PW LS 8997 with an emulsion and hydrogel containing 0.1%.



A Beauty Creations Consumer Study: stressful emotions are perceived to negatively affect the skin, leading to dull and dry skin

Mindfulness is full awareness of the experience of the present moment. It is strongly correlated with greater well-being and perceived health **Mindfulness** is used to reduce stress and anxiety and to manage our emotions. In recent studies in the press people have shown that mindful people were more resilient and better able to manage stressful situations.

OBJECTIVE

Mindfulness is a meditation technique that is said to help alleviate stress which would have a positive impact on a person's appearance. To better understand how consumer, relate to the concept of mindfulness, Beauty Creations conducted a large consumer survey regarding psychological stress and its effects upon the appearance of consumers, and particularly on their skin. This study was done on one hundred female volunteers aged between 25 and 65 years, that completed an online consumer survey.

RESULTS AND DISCUSSION

Respondents reported that they routinely experience stress in their daily lives. Respondents' open-ended comments indicated that the experience of stress leads to a general tendency to not take care of one's appearance as one normally would. Approximately 40% of the study population, regardless of age group, was familiar with the concept of Mindfulness and many of those who were familiar with it tried to practice some elements of it in their daily lives. When asked to explain the concept of Mindfulness in their own terms, most respondents described it as being self-aware. This idea included focusing on tasks at hand, appreciating moments as they come or being conscious of and accountable for decisions as they are made. Many respondents freely associated Mindfulness with stress alleviation.

Most respondents did report a direct effect on their appearance due to stress.

A majority of consumers, 90% agreed that the stress negatively affects their skin (Figure 2).

More precisely, most respondents also felt they experienced dullness or lack of radiance (73%) as a reaction to stress in their lives. To a lesser extent, 39% of the women also perceived a visible decrease in hydration of their skin when they experienced stress (Figure 3).



Figure 2 - Psychological stress negatively affects the skin (consumer survey: 100 female volunteers - 25 - 65 years old).

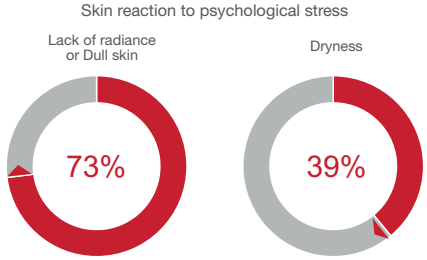


Figure 3 - Skin reaction to psychological stress (consumer survey: 100 female volunteers - 25 - 65 years old).

CONCLUSION

As demonstrated by this consumer survey targeting the impacts of stressful situations on skin appearance, psychological stress is perceived by consumers to be a factor negatively affecting the appearance of their skin, in particular, the reported dull skin with lack of radiance and increased dryness.

MATERIALS&METHODS

One hundred American female subjects between 25 and 65 years old participated in this study.

Respondents received a link to the consumer survey via email and completed the questionnaire at home. The survey included a combination of closed-ended quantitative questions and open-ended qualitative questions that allowed the respondents to answer in their own words according to their personal experiences and interpretation of the question.



A DEMONSTRATED EFFICACY

Hyalurosmooth™ has been proven to minimize visual the side effects of cortisol produced by the accumulation of stress in skin. The skin is rebalanced in a mindful environment.

Hyalurosmooth™ was evaluated through *in vitro*, *ex vivo* and *in vivo* testing on the well-known main stress mediator affecting collagen and hyaluronic acid, which results in skin dryness and dull skin.

Epidermal cortisol modulation by acting on:
- Beta11HSD1 enzyme decrease.
- Beta11HSD2 enzyme increase.

Dermal protection against cortisol's negative effects by acting on:
- Collagen type I.
- Hyaluronic acid.

Proven *ex vivo* and *in vivo* performance:
- *Ex vivo*, for its quick moisturizing activity on *stratum corneum*.
- *In vivo*, for its immediate effect on skin complexion.
- *In vivo*, for its mindful effect on skin moisturization and radiance on volunteers affected by a stressful life.

EFFICACY

Controlling cortisol production by keratinocytes

Psychological stress affects skin's hydration and radiance and cortisol is involved in this pathway. Cortisol circulation is triggered by psychological stress to increase the ability of the body to react quickly to stressors. Accumulation of stressful situations (with high levels of cortisol) induces adverse effects on the skin such as a decrease of collagen or hyaluronic acid.

Local cortisol production in the skin has been described through 11β-hydroxysteroid dehydrogenases (11β-HSDs) via a de novo pathway[23]. The balance of the two isoforms 11β-HSD1 (HSD1) and 11β-HSD2 (HSD2) catalyzes the interconversion between cortisone and cortisol (Figure 4).

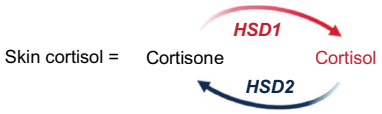


Figure 4 - Illustration of interconversion between cortisone and cortisol[24].

in vitro

OBJECTIVE

The purpose of this study was to monitor the effect of Hyalurosmooth™ on the HSD1 and HSD2 enzymes.

The test was done on primary human keratinocytes. The impact of Hyalurosmooth™ was evaluated by qRT-PCR.

This study was conducted with the powder form of Hyalurosmooth™ (PW LS 8997).

RESULTS & DISCUSSION

In figure 5A results showed that Hyalurosmooth™ at 0.05% significantly decreased the HSD1 (enzyme favorizing cortisol production) expression by 61% in keratinocytes in a dose-dependent manner ($p<0.01$). Inversely, Hyalurosmooth™ increased the expression of HSD2 (enzyme favorizing cortisol reduction) by 215% at the same concentration (figure 5B).

CONCLUSION

Hyalurosmooth™ helps to modulate the cortisol level in the skin by decreasing the expression of HSD1 and increasing the expression of HSD2.

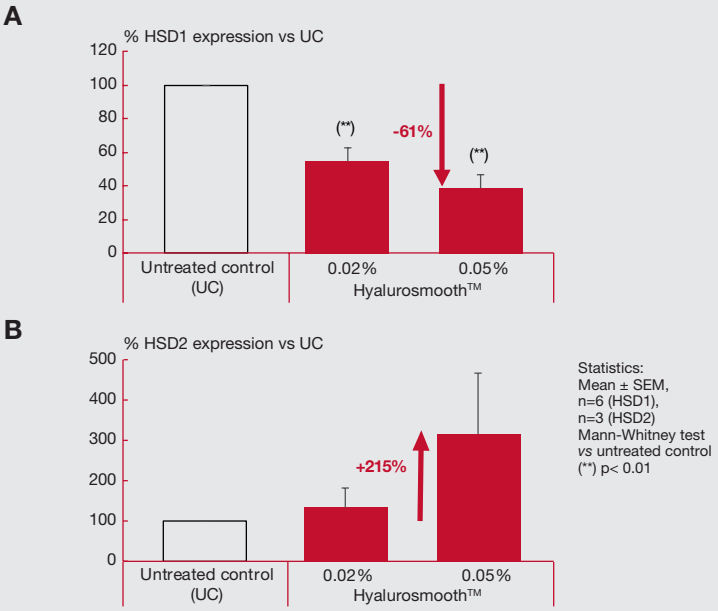


Figure 5 - (**A**) HSD1 expression after 72h of Hyalurosmooth™ (PW LS 8997) treatment. (**B**) HSD2 expression after 72h of Hyalurosmooth™ (PW LS 8997) treatment.

MATERIALS&METHODS

Cell culture and treatments

Human normal epidermal keratinocytes were isolated by trypsinization from abdominal biopsies of female donors between 37- and 55-year-old obtained from plastic surgery. Keratinocytes were cultured in a DMEM medium (Dulbecco's Modified Eagle Medium) containing 10% FBS (fetal bovine serum) and antibiotics. Then, they were incubated at 37°C in the presence of 5% CO_2 for 3 to 4 days until confluence.

Hyalurosmooth™ PW LS 8997 was added at 0.02% and 0.05% in the DMEM without FBS in cultures. The cells were incubated at 37°C in the presence of 5% CO_2 for 72h, then cell pellets were frozen at -80°C.

Real time RT-PCR of HSD1 and HSD2

Total RNAs were extracted from the different pellets (NucleoSpin® 96 RNA kit, Macherey-Nagel), then quantified by measuring the optical densities at 260 and 280 nm. These measures also allow validation of the RNA quality. The reverse transcription (RT) step was done on 0.2 to 2 µg of RNAs from the different tested conditions, depending on the gene being tested. RT was performed in a 20 µl final volume, as described in the manufacturer instructions (Thermoscript™ RT-PCR System, Invitrogen).

The quantitative PCRs (qPCR) to analyze the target gene HSD11B1 and housekeeping genes EEF1A1 and SDHA were carried out on 2 µl of reverse transcription reaction mix diluted at 1/2 using specific oligonucleotides and LC 480 SYBR Green I Master kit (Roche) on a LightCycler® 480 thermocycler (Roche). Each assay was analyzed twice in duplicate. To analyze the HSD11B2 target gene and the housekeeping genes SDHA and ACTB, the qPCRs were carried out on 2 µl of the RT reaction mix diluted at 1/20 using specific oligonucleotides. Each assay was analyzed in duplicate. For HSD11B1 and HSD11B2 gene analysis, the SYBR Green fluorescence was measured after each amplification cycle. Fusion curves were performed after qPCR to validate amplification specificity. The relative quantification was performed using the ΔΔCt method after analysis of calibration curves. The results come from three independent cultures. They were expressed as percentage of gene expression compared to the non-treated condition as 100%.

Statistics

The statistical analysis was carried out using the Mann-Whitney test vs untreated control. The significance threshold was fixed at 0.05 for the p-value (**) $p<0.01$ vs cortisol stress.

EFFICACY

Prevention of collagen type I decrease by cortisol stress in fibroblasts

in vitro

OBJECTIVE

Psychological stress affects the cortisol level and induces adverse effects on the skin such as decreasing the collagen type I level in fibroblasts. The goal of this study was to monitor the effect of Hyalurosmooth™ on collagen type I synthesis in fibroblasts treated with cortisol. This study was performed on human primary fibroblasts by ELISA assay.
This study was conducted with the powder form of Hyalurosmooth™ (PW LS 8997).

RESULTS & DISCUSSION

When cells are incubated with cortisol, we observed a significant ($p<0.001$) decrease of collagen type I synthesis. Addition of Hyalurosmooth™ at 0.05% has allowed us to significantly counterbalance by 13% ($p<0.05$) the cortisol-induced decrease of collagen type I (Figure 6).

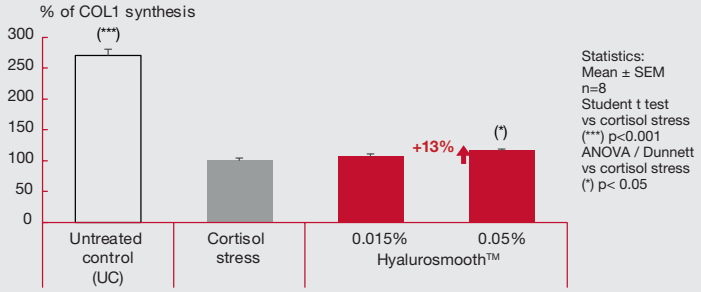


Figure 6 - Effect of Hyalurosmooth™ (PW LS 8997) on collagen type I synthesis by fibroblasts under cortisol stress.

CONCLUSION

Hyalurosmooth™ helps to reverse cortisol's inhibitory effect on collagen by increasing the collagen type I synthesis.

MATERIALS&METHODS

Cell culture and treatments

Human dermal fibroblasts from adult donors (women 50 and 66 years-old) were seeded at 30 000 cells per cm² and incubated within DMEM growth medium (Dulbecco's Modified Eagle's Medium, Invitrogen France) with 10% Fetal Calf Serum (DUTSCHER France) for 24 hours at 37°C. Then, the cells were treated with the addition of cortisol (Sigma) and with or without the product (Hyalurosmooth™ PW LS 8997) at 0.015 and 0.05%.

ELISA method

After a second incubation of 72 hours at 37°C, the level of type I collagen released was determined by measuring the amount of type I collagen in culture medium through an ELISA method according to the recommendations of the manufacturer (Kit Metra™ CICP EIA TecoMedical). The results were calculated by referring to a range of collagen type I concentrations and then presented as a percentage referring to the cortisol state.

Statistics

The statistical evaluation is based on Student t-test or one way ANOVA vs cortisol stress, according to Sigmaplot™ evaluation. The significance threshold was fixed at a p-value of (*) $p<0.05$ vs cortisol stress.

EFFICACY

Restoration of hyaluronic acid synthesis decreased by cortisol stress in fibroblasts

in vitro

OBJECTIVE

Psychological stress affects the cortisol level and induces adverse effects on the skin such as decreasing the level of hyaluronic acid in fibroblasts.
The goal of this study was to monitor the effect of Hyalurosmooth™ on hyaluronic acid synthesis in fibroblasts treated by cortisol.
This study was performed on human primary fibroblasts using the ELISA assay.
This study was conducted with the powder form of Hyalurosmooth™ (PW LS 8997).

RESULTS & DISCUSSION

When cells are incubated with cortisol, we observed a significant ($p<0.05$) decrease of hyaluronic acid synthesis.

Figure 7 shows that the addition of Hyalurosmooth™ allowed to significantly counterbalance by +99% at 0.015% ($p<0.05$) the cortisol-induced decrease of hyaluronic acid.

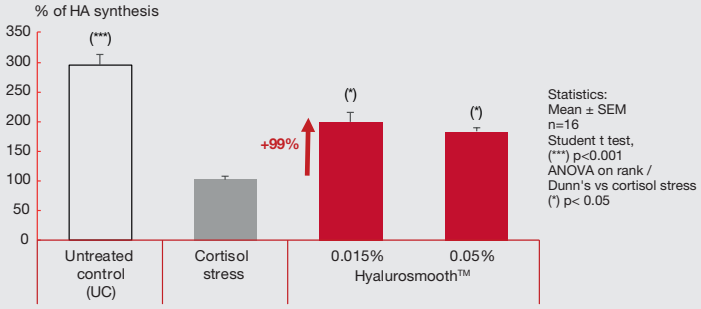


Figure 7 - Effect of Hyalurosmooth™ (PW LS 8997) on hyaluronic acid synthesis by fibroblasts under cortisol stress.

CONCLUSION

Hyalurosmooth™ helps to reverse cortisol's inhibitory effect on hyaluronic acid by increasing hyaluronic acid synthesis.

MATERIALS&METHODS

Cell culture and treatments

Human dermal fibroblasts from adult donors (women of 41, 50 and 66 years-old) were seeded at 30 000 cells per cm^2 and incubated within DMEM growth medium (Dulbecco's Modified Eagle's, Invitrogen France) with 10% Fetal Calf Serum (DUTSCHER France) for 24 hours at 37°C. Then, the growth medium was replaced by a standard medium containing TGFb1 (Transforming Growth Factor-b1 Sigma T-7039) with addition of cortisol (Sigma) and with or without the product (Hyalurosmooth™ PW LS 8997) at 0.015 and 0.05%.

ELISA method

After an incubation of 72 hours at 37°C, the level of hyaluronic acid released in the culture medium was determined by an ELISA method according to the recommendations of the manufacturer (kit Echelon K-1200, Tebu Bio, France). The results were calculated referring to a range of hyaluronic acid concentrations and then presented as a percentage referring to the cortisol state.

Statistics

The statistical evaluation is based on Student t-Test or ANOVA on rank vs cortisol stress, according to Sigmaplot™ evaluation. The significance threshold was fixed at a p-value of (*) $p<0.05$ vs cortisol stress.

EFFICACY

Immediate moisturizing activity on human *stratum corneum*

ex vivo

OBJECTIVE

The goal of this study was to evaluate the moisturizing activity of Hyalurosmooth™#* on the *stratum corneum* compared to Sodium Hyaluronate, by measuring the dielectric conductivity. Both products were formulated in a hydrogel and compared to placebo hydrogel.
This study was conducted with the liquid form of Hyalurosmooth™#* (LS 8998).

RESULTS & DISCUSSION

The application of hydrogels containing Hyalurosmooth™#* and sodium (Na) hyaluronate have induced an increase of dielectric conductivity of the *stratum corneum*, indicating a moisturizing effect. At 0.2%, but not 0.07%, Na hyaluronate significantly increased the dielectric conductivity vs the placebo hydrogel for up to 6 hours (Fig 8A). After 24 hours, Na hyaluronate did not display any significant change of dielectric conductivity compared to the placebo.

With Hyalurosmooth™ at 5% (liquid form LS 8998, equivalent to 0.1% of the powder form PW LS 8997), the dielectric conductivity significantly increased compared to the placebo for up to 24 hours (Fig 8B). It outperformed the placebo by 44% after 30 min ($p<0.01$) and by 23.5% after 24 hours ($p<0.05$).

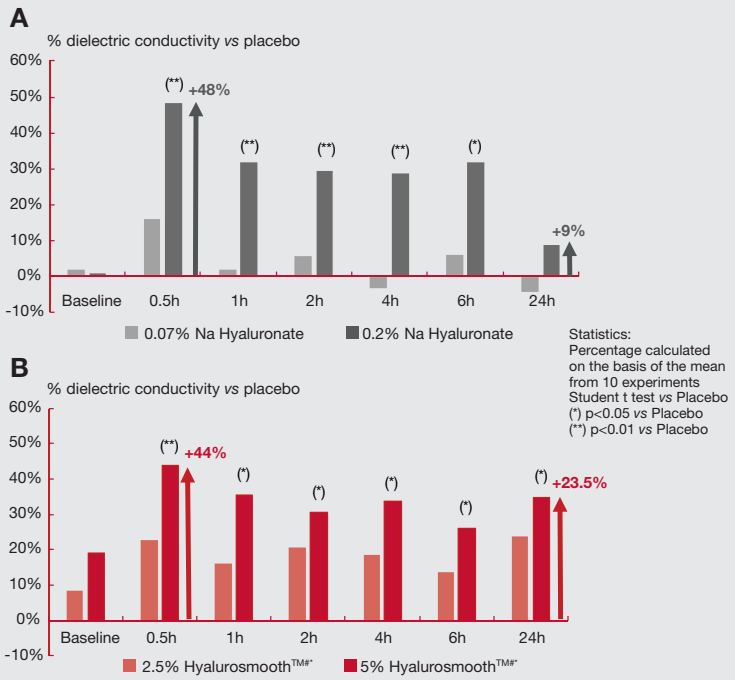


Figure 8 - effect of Na Hyaluronate (**A**) and Hyalurosmooth™#* (LS 8998) (**B**) on dielectric conductivity in the human *stratum corneum*. Only significant results vs placebo are indicated on the graph.

CONCLUSION

At similar concentrations[2] of active matter, Hyalurosmooth™#* provided moisturization equivalent to the benchmark, sodium hyaluronate, up to 6 hours. Moreover, Hyalurosmooth™#* out performed the benchmark with respect to the long-term effects and gave significant 24 hour moisturization.

[2] Study conducted with 0.2% of sodium hyaluronate and 0.1% of Cassia Angustifolia Seed Polysaccharides (equivalent to 5% Hyalurosmooth™ LS 8998 and 0.1% Hyalurosmooth™ PW LS 8997)
The dielectric conductivity study was evaluated with Cassia angustifolia extract at a dose equivalent to Hyalurosmooth™ PW at 0.05% and 0.1%.
** This product does not contain hyaluronic acid (derivatives)*

ex vivo

MATERIALS&METHODS

Study design

Evaluation of the moisturizing activity on stratum corneum *ex vivo* by dielectric conductivity measurement, compared to a placebo hydrogel and a control without treatment.

Method

The epidermis was isolated from surgical skin biopsies. Then, the *stratum corneum* (horny layer) was separated from the living layers of the epidermis, cut into small pieces and put onto filter paper previously dampened with PBS. The *stratum corneum* and filter paper were deposited onto a microscope slide and fixed in position by adhesive tape (figure 9).

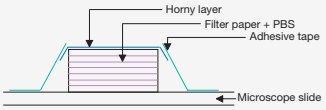


Figure 9 - Illustration of the moisturizing study on the horny layer by dielectric conductivity.

The conductivity measurements were carried out by Skincon 200 after stabilization at 20°C and 44% relative humidity (RH) according the protocol in Figure 10.

The hydrogels were prepared as water solutions of the products to be tested with 0.40% carbomer and preservative.

Six different conditions were tested:
- horny layer without treatment,
- horny layer treated with a hydrogel placebo,
- horny layer treated with a hydrogel with 0.07% Na hyaluronate,
- horny layer treated with a hydrogel with 0.20% Na hyaluronate,
- horny layer treated with a hydrogel with 2.5% Hyalurosmooth™#* (LS 8998),
- horny layer treated with a hydrogel with 5% Hyalurosmooth™#* (LS 8998).

The dielectric conductivity was measured up to 24 hours after treatment.

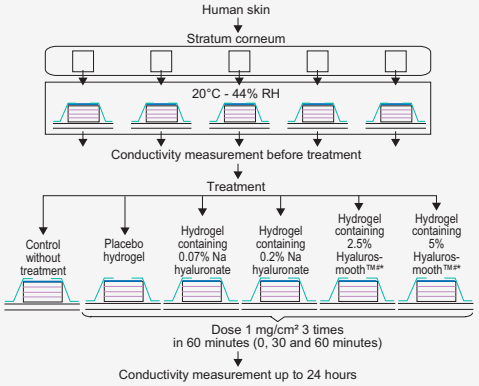


Figure 10 - Schematic protocol of the conductivity measurements carried out by the Skincon 200 system.

Statistics

The statistical evaluation is based on Student's t-Test vs Placebo according to Sigmaplot™ evaluation. The significance threshold was fixed at a p-value of 0.05. (*) $p<0.05$ *vs* Placebo, (**) $p<0.01$ *vs* Placebo.

Conclusions *in vitro, ex vivo*

By acting on the main stress mediator, Hyalurosmooth™, preserves the skin in a mindful state.

In vitro, Hyalurosmooth™ has demonstrated its efficacy on key mindful-related targets affected by the stress mediator in the epidermis and dermis.

Balances epidermal cortisol by acting on:
- Beta11HSD1 enzyme: 61% decrease.
- Beta11HSD2 enzyme: 215% increase.

Cortisol's side effect minimization by acting on:
- Collagen type I 13% increase.
- Hyaluronic acid 99% increase.

Quickly moisturizes the *stratum corneum* by acting on:
- Conductivity increase at 30 min by 44%.
- Conductivity increase at 24 hours by 23.5%.

EFFICACY

Improves radiance and hydration of the skin affected by accumulation of stressful situations

in vivo

OBJECTIVE

In our previously described consumer survey, we showed that psychological stress is perceived by consumers to affect their skin hydration and radiance.

In this double-blind placebo controlled study, we evaluated the ability of Hyalurosmooth™ to improve these major signs of stressed skin on volunteers with self-perceived stress experienced in life.

This study was conducted with the powder form of Hyalurosmooth™ (PW LS 8997).

Structural and molecular alterations of the skin, in particular, collagen and hyaluronan degradation, have an impact on the skin's radiance and hydration level. The effects of Hyalurosmooth™ (powder form, PW LS 8997) at 0.1% on the radiance and hydration level of the skin were measured respectively by Glossymeter (instrumental radiance evaluation) and Skicon (instrumental moisturization evaluation). Measurements were taken at baseline and 28-day time points compared to the placebo (Figure 11).

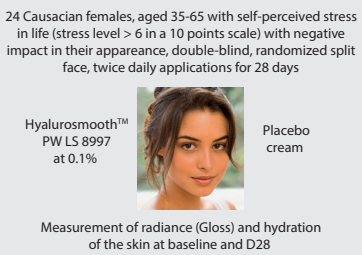


Figure 11 - Face radiance (Gloss) and hydration clinical study on volunteers affected by a stressful life, Hyalurosmooth™ vs placebo.

RESULTS & DISCUSSION

Hydration improvement on the skin of volunteers affected by a stressful life

On people with dry skin and a perceived stressful life, Hyalurosmooth™ significantly improved (by 37%) skin hydration after 28 days vs the baseline (Figure 12), whereas the Placebo did not display any significant effect.

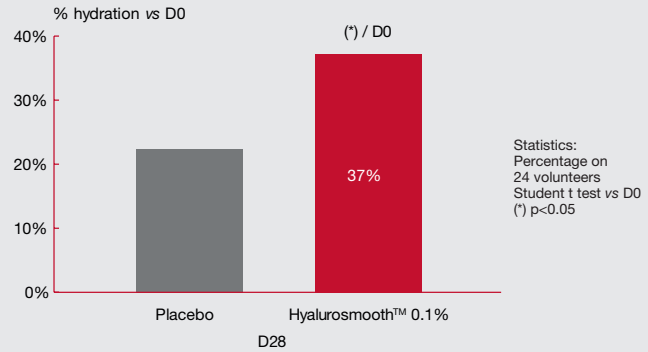


Figure 12 - Quantitative improvement of Hyalurosmooth™ (PW LS 8997) treatment vs D0 measured by Instrumental evaluation of skin hydration (Skicon).

Radiance improvement on the skin of volunteers affected by a stressful life

On people with dry skin and a perceived stressful life, Hyalurosmooth™ significantly, improved (by 16%) skin radiance (Gloss) after 28 days vs the baseline (Figure 13), whereas the placebo did not display any significant effect.

Gloss is defined as uneven texture of the skin and represents the skin's extrinsic radiance.

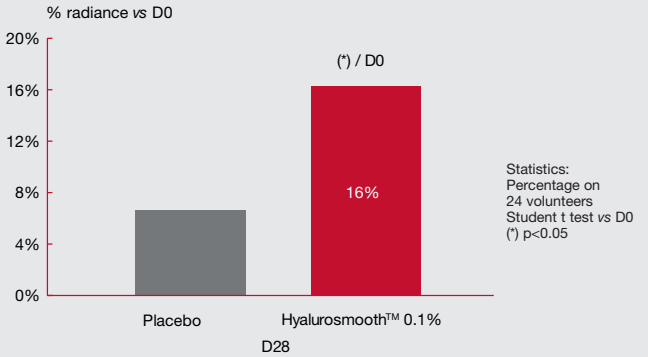


Figure 13 - Quantitative improvement of Hyalurosmooth™ (PW LS 8997) treatment vs D0 measured by Instrumental evaluation of skin radiance (Glossymeter).

CONCLUSION

Hyalurosmooth™ demonstrated its ability to mimic the positive effects of a "Mindful" way of life on the skin by improving skin hydration and radiance, specifically on volunteers affected by a stressful life.

in vivo

MATERIALS&METHODS

Study design

The clinical study was conducted on 24 healthy female volunteers affected by perceived stressful life. It was carried out as a placebo-controlled double-blind randomized, split-face study. The mindful-like effect on skin was assessed by the measurement of skin hydration and radiance (Gloss) at baseline and after 28 days of treatment with Hyalurosmooth™ at 0.1%. The formulations are detailed in Annex 2.

Inclusion criteria

Female volunteers, from 35 to 65 years old, with dry skin and self-perceived experiences of stress in life, specifically at work, in family and with financial problems.
They were evaluated at a stress level > 6 on a 10-point scale. Moreover, they agreed that the stress they experience has a negative impact in their skin's appearance.

Application modality

The products (a formulation containing Hyalurosmooth™ (PW LS 8997) at 0.1% or a placebo formulation) were applied by the volunteers twice a day on each half face for 28 days, under the normal conditions of use.

Evaluation methods

Skicon (instrumental moisturization tool)
This technique allows the evaluation of the moisturization on the *stratum corneum* (10-20 mm) and is based on the measurement of the conductance of the skin. Higher Skicon values are associated with greater hydration.

Glossymeter (instrumental radiance evaluation)
Gloss is defined as uneven texture of the skin. The measurement is based on the "Reflect Light" Technology of the skin Surface. This technique allows the skin to be uniformity visualized. The higher the skin evenness, the higher the gloss rate and the higher the proper skin reflection is. Higher values are associated with greater radiance (less dull skin).

Statistics

The statistical evaluation is based on Student t test *vs* D0 according to Sigmaplot™ evaluation. The significance threshold was fixed at a p-value of (*) $p<0.05$.

EFFICACY

Immediate improvement of skin complexion

in vivo

OBJECTIVE

Skin complexion is a global parameter. Its evaluation requires the simultaneous consideration of skin clarity, color, evenness and texture. All these properties contribute to a good appearance, i.e. smooth skin and a radiant complexion.

In this double-blind randomized study, the perception of the benefits on skin complexion of a cream containing 3% Hyalurosmooth™$* was evaluated on the face, compared to a cream containing 0.06% sodium (Na) hyaluronate (benchmark). Twelve trained specialized were used to rank skin complexion before and 3, 15 and 30 minutes after a single split face application (Figure 14). The evaluation was done according to a 7-point scale (1=not at all to 7=very intense). This study was conducted with the liquid form of Hyalurosmooth™$* (LS 8998).

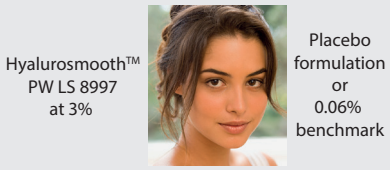


Figure 14 - Clinical study on volunteers for the evaluation of the immediate effect of Hyalurosmooth™$* on the skin complexion vs placebo or Benchmark.

RESULTS & DISCUSSION

The results are presented Figure 15. With the cream containing Hyalurosmooth™$* at 3%, the panelists observed an immediate visible improvement of skin complexion *vs* baseline. Hyalurosmooth™ significantly improved skin complexion by 47% and 51% at 15 and 30 minutes respectively vs placebo ($p<0.01$). Moreover, the effect on complexion of Hyalurosmooth™$* is significantly better than the placebo at all time points ($p<0.01$), and significantly better than sodium hyaluronate 15 and 30 minutes after application ($p<0.01$). Hyalurosmooth™$* outperformed the placebo and benchmark approximately three-fold.

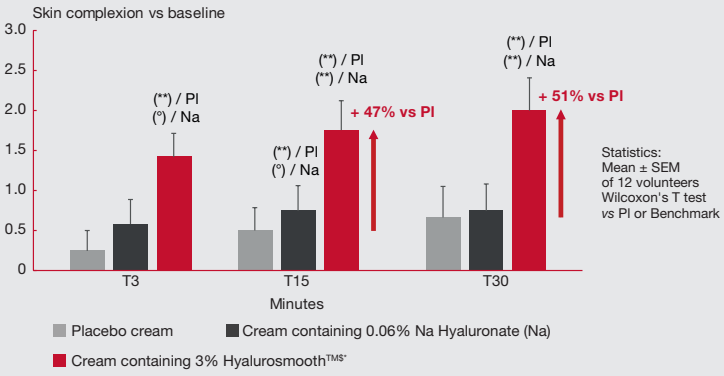


Figure 15 - Sensorial evaluation of skin complexion by trained volunteers. Hyalurosmooth™$* (LS 8998) at 3% vs sodium hyaluronate and vs placebo).

CONCLUSION

Compared to placebo and to sodium hyaluronate, Hyalurosmooth™$* visibly and significantly improved skin complexion immediately.

$ The skin complexion study was evaluated with Cassia angustifolia extract at a dose equivalent to Hyalurosmooth™ PW at 0.06%.
** This product does not contain hyaluronic acid (derivatives)*

MATERIALS&METHODS

Study design
The clinical study was carried out as a placebo-controlled double-blind randomized study on the face for the evaluation of skin complexion after treatment with Hyalurosmooth™$* vs sodium hyaluronate. The formulations are detailed in Annex 2.

Inclusion criteria
Twelve trained specialized were used to describe and assess the sensory efficacy of cosmetic products, according to standardized guidelines.

Application modality
The products (formulation containing Hyalurosmooth™$* (LS 8998) at 3% or sodium hyaluronate at 0.06% or placebo formulation were applied by the trained volunteers as a single application by on areas of the face, in standardized environmental conditions defined by AFNOR standards.

Evaluation methods
The evaluation was monadic with, each trained specialized describing the first product's effect on skin complexion, before evaluating the second product.

The skin complexion was evaluated:
- before application,
- 3, 15 and 30 minutes after application of each product.
The evaluation of skin complexion was done according to a scale from 1=not at all to 7=very intense.

Statistics
The mean, SEM (standard error of mean) were calculated. Wilcoxon's T test on paired series was used to compare the data vs placebo or benchmark.

Conclusions *in vivo*

Hyalurosmooth™, helps to rebalance a mindful skin environment through a botanical alternative to hyaluronic acid.

In vivo, Hyalurosmooth™, has demonstrated both efficacy on key mindful-related end points affected by accumulation stressors and an immediate effect on skin complexion:

Hydration recovery: 37% improvement.

Radiance recovery: 16% improvement.

Immediate complexion: 51% improvement.



GENERAL CONCLUSION

Hyalurosmooth™: the mindful way to cope with everyday stress

When you have a stress life, specifically at work or in family life with financial worries, is perceived to have negative impact on the appearance.

To better understand this negative impact, BBCS has designed a consumer survey of 100 women consumers. Interestingly, two cutaneous aspects were mainly perceived to be affected by stressful life: lack of radiance and increased dryness. In addition, several authors mentioned that psychological stress via cortisol release, can affect the skin at the epidermis and dermis levels by decreasing structural elements.

To counteract the negative impact of psychological stress, the practice of **mindfulness** can be helpful. It has been demonstrated that its practice is strongly correlated with greater **well-being** and better health.

In looking for mindful care, we have designed a solution with Hyalurosmooth™, natural galactomannans from the seed of the Ayurveda plant *Cassia angustifolia*.

In vitro, Hyalurosmooth™ demonstrated its ability to modulate two enzymes, beta11-HSD1 and beta11-HSD2, that affect the cortisol level in the epidermis. Hyalurosmooth™ also preserved the structural cutaneous elements. It helped to preserve collagen type I and hyaluronic acid.

Moreover, *ex vivo*, on *stratum corneum*, Hyalurosmooth™ provided an efficient moisturization effect at 6 hours equivalent to the benchmark (sodium hyaluronate). Hyalurosmooth™ outperformed the benchmark with a long-term and significant moisturizing effect at 24 hours.

In vivo, tested in a placebo-controlled study on a panel with a stressful life and dry skin, Hyalurosmooth™ demonstrated its ability to specifically counteract the adverse effects of stress on skin by improving skin hydration and radiance. **Hyalurosmooth™ was able to restore mindful skin**. Hyalurosmooth™ also improved immediate skin complexion and demonstrated a better efficiency than the sodium hyaluronate.

Hyalurosmooth™ is a *Cassia angustifolia* extract rich in galactomannan completely devoid of hyaluronic acid while showing similar functions. Due to its ability to specifically counteract the adverse effects of stress, **Hyalurosmooth™ restores a mindful skin, increasing hydration and radiance in stressed people**.

ANNEX 1

Technical data - Available upon request

- Quality and Regulatory Product Information
- Composition sheet
- Specifications
- Formulation Data Sheet

ANNEX 2

Clinical test formula for skin hydration and radiance

Trade name	INCI name	Placebo formulation %	Hyalurosmooth™ formulation %
Water	Water	sqf 100	sqf 100
Cetiol® Sensoft	Propylhetyl Caprylate	5.00	5.00
Cetiol® CC	Dicaprylyl Carbonate	5.00	5.00
Cetiol® OE	Dicaprylyl Ether	5.00	5.00
Eumulgin® S21	Steareth-21	2.50	2.50
Elestab™ 388	Propylene Glycol (and) Phenoxyethanol (and) Chlorphenesin (and) Methylparaben	2.50	2.50
Eumulgin® S2	Steareth-2	2.00	2.00
Cosmedia® ACE	Sodium Polyacrylate (and) Dicaprylyl Carbonate (and) Polyglyceryl-3 Caprate	1.48	1.48
Lanette® O	Cetearyl Alcohol	1.00	1.00
Rheocare® XGN	Xanthan Gum	0.20	0.20
Hyalurosmooth™ LS 8997	*Cassia Angustifolia* Seed Polysaccharide	-	0.10
Edeta® BD	Disodium EDTA	0.05	0.05

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Clinical test formula for Immediate improvement of skin complexion

Trade name	INCI name	Placebo formulation %	Hyalurosmooth™ formulation %	Benchmark 0.06% formulation %
Water	Water	sqf 100	sqf 100	sqf 100
	Preservative	sq	sq	sq
Emulgade® 1000 NI	Cetearyl Alcohol (and) Ceteareth 20	5.00	5.00	5.00
Eutanol® G	Octyldodecanol	5.00	5.00	5.00
Emulgade® 165	Glyceryl Stearate (and) PEG-100 Stereate	3.50	3.50	3.50
Hyalurosmooth™ LS 8998	Water (and) Cassia Angustifolia Seed Polysaccharide (and) Hydroxyethylcellulose	-	3.00[a]	-
	Mineral Oil	3.00	3.00	3.00
Cutina® GMS V	Glyceryl Stearate	2.50	2.50	2.50
Lanette® O	Cetearyl Alcohol	2.00	2.00	2.00
	Sodium Hyaluronate	-	-	0.06

[a] Equivalent to 0.06% Hyalurosmooth™ PW LS 8997

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ANNEX 3

Formulation examples

Soothing silky cream (face care)

Phase	Ingredients	INCI	% by weight	Function
A	Emulgade® PL 68/50	Cetearyl Glucoside, Cetearyl Alcohol	2.00	Emulsifier (O/W)
	Cutina® HVG	Hydrogenated Vegetable Glycerides	1.50	Consistency agent
	Cetiol® Sensoft	Propylheptyl Caprylate	5.00	Emollient
	Cegesoft® PS 6	Olus Oil [EU], Vegetable Oil [CTFA]	3.00	Emollient
	Cetiol® Ultimate	Undecane, Tridecane	4.00	Emollient
	Cetiol® 4 All	Dipropylheptyl Carbonate	4.00	Emollient
	Lipofructyl™ Argan LS 9779	Argania Spinosa Kernel Oil	1.00	Emollient
	Chione™ M SVA	Synthetic Fluorphlogopite, Lauroyl Lysine	2.00	Skin feel modifier
	Xiameter PMX-200 Silicone Fluid 50CS *(Dow Corning)*	Dimethicone	2.00	Defoamer
	Cosmedia® SP	Sodium Polyacrylate	0.60	Rheology modifier
B	Water, demin.	Aqua	70.20	
	Glycerin	Glycerin	4.00	Humectant
	Eumulgin® SG	Sodium Stearoyl Glutamate	0.40	Emulsifier (O/W)
	Preservative		q.s.	Preservative
	Hyalurosmooth™ PW LS 8997	Cassia Angustifolia Polysaccharides	0.10	Active ingredient
C	Perfume*	Parfum	0.20	Fragrance

Specifications	
pH value (23°C)	6.3
Viscosity (Brookfield; RVT; spindle TC, helipath; 20 rpm; 23°C)	50000 mPas

Disclaimer

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Face delight cream (face care)

Phase	Ingredients	INCI	% by weight	Function
A	Emulgade® Sucro Plus	Sucrose Polystearate, Cetyl Palmitate	1.00	Emulsifier (O/W)
	Cetiol® LC	Coco-Caprylate/Caprate	7.50	Emollient
	Cetiol® Sensoft	Propylheptyl Caprylate	2.50	Emollient
	Cetiol® 4 All	Dipropylheptyl Carbonate	2.00	Emollient
	Lanette® O	Cetearyl Alcohol	4.00	Consistency agent
B	Water, demin.	Aqua	35.80	
	Rheocare® C Plus	Carbomer	0.20	Rheology modifier
C	Water, demin.	Aqua	42.82	
	Glycerin	Glycerin	3.53	Humectant
	Preservative		q.s.	Preservative
	Hyalurosmooth™ PW LS 8997	Cassia Angustifolia Polysaccharides	0.10	Active ingredient
D	Puricolor® Red ARE33 FDA (1% solution)	CI 17200	q.s.	Colorant
	Perfume*	Parfum	0.25	Fragrance
	Potassium Hydroxide (20% solution)	Potassium Hydroxide	0.30	pH Adjustment

Specifications	
pH value (23°C)	6,25
Viscosity (Brookfield; RVT; spindle TD, helipath; 5 rpm; 23°C)	72 000 mPas

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